

BENNETT UNIVERSITY

School of Computer Science Engineering and Technology

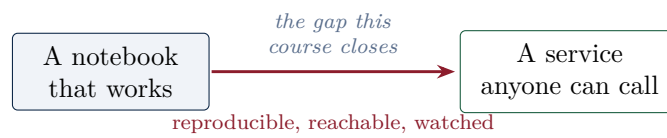
B.Tech Computer Science and Engineering, Semester V

SCSE3040 • Machine Learning Operations

Lecture 01

Why MLOps

From a notebook that runs once,
to a service that runs every day



Lecture content only. Cross checked against the approved Teaching Learning and Assessment Plan, session plan Sessions 1 and 2 of Module I. Course policy, assessment rubrics and laboratory setup are in the separate Course Handout and Laboratory Manual, and nothing is duplicated between them.

Parts I to IV are the taught hour. Parts V and VI are a structured week of self study, with every resource verified.

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How this handout maps to the approved session plan

The BTLAP splits this material across two lecture sessions. Your lecture group meets twice a week, so the natural division is Session 1 in the first slot and Session 2 in the second. This handout covers both, and every topic listed in the session plan is accounted for below.

Session	Topic in the approved session plan	Where it is covered here
1	Overview of MLOps	Part I, Sections 1 to 4
1	Semester running project methodology, notebook to production	Part I, Section 8
1	Discussion: where do machine learning projects fail after the notebook stage	Part I, Sections 5 to 7, and the paired exercise in Part IV
1	Course outcomes, evaluation components, CLA rubric	<i>Deliberately not here.</i> These are in the Course Handout, which is the document the session plan names as the reading
2	History and evolution of MLOps	Part II, Section 9
2	Core principles and lifecycle of production machine learning	Part II, Sections 10 and 11
2	DevOps to MLOps	Part II, Sections 13 and 14
2	Bridging Data Science, DevOps and Engineering roles	Part II, Section 12
2	Activity: contrast a Kaggle notebook with a bank's fraud detection service	Part III, a worksheet with a grid to complete in class

The taught hour, minute by minute

0 to 8	Section 1, the story. Ask the two opening questions before showing anything
8 to 16	Sections 2 and 3, notebook against production, and what production demands
16 to 24	Section 4, the three properties, with the quiet failure box on the board
24 to 38	Section 6, two of the four cases. Zillow and one other. Do not rush these
38 to 46	Section 5, the four exits, drawn live as students name failures
46 to 54	Section 10, the lifecycle as a loop, and Section 14, three moving parts
54 to 60	Section 8, the running project, and the closing frame
Sections 9, 11, 12 and 13 are the second slot. Parts V and VI are never lectured; they are released for the week.	

PART I

Why MLOps

1 A model that worked, and still failed

Four students build a delivery time predictor for a food delivery application. They clean the data, engineer distance and time of day features, train a gradient boosted model, and reach a mean absolute error of four minutes on held out data. The demonstration gets applause. A regional restaurant chain asks whether they can start using it from Monday.

Within a week, three things surface.

1. **It runs on one laptop.** The chain's ordering application cannot call it, because there is no address to call. The model exists only as an object in memory inside a notebook session.
2. **It needs a file in a Downloads folder.** The training data was loaded from a local path. Nobody else can rebuild the model, because nobody else has that exact file, and nobody recorded which version of it was used.
3. **Three months later it is quietly wrong.** Two new delivery zones opened, fuel prices changed rider behaviour, and the predictions drifted. No error appeared. No alert fired. Customers simply began receiving times that were consistently optimistic.

None of these is a modelling problem. The model was fine. Every one of them is an engineering problem, and all three are solved by material in this course.

The distinction to hold on to

Cooking one dish at home is not the same as running a kitchen for five hundred plates a night. Same recipe, completely different demands: consistency, speed, staffing, and a plan for when something goes wrong.

2 A notebook, and a production service

A NOTEBOOK

Runs when *you* press Shift and Enter
One user, and that user is you
Breaks silently, and only you notice
Data sits in a folder on your disk
Success is a number in a cell

Optimised for exploration. The correct tool for finding out whether an idea works at all.

A PRODUCTION SERVICE

Runs on its own, day and night
Many users and systems, at once
Must be watched, and must alert somebody
Data keeps arriving and keeps changing
Success is a decision somebody trusted

Optimised for reliability. The correct tool for delivering the idea to people who are not you.

Figure 1: The two are not competitors. The notebook is where the idea is found; the service is where the idea is delivered. Almost every difficulty in this course comes from moving something between them.

3 What production actually demands

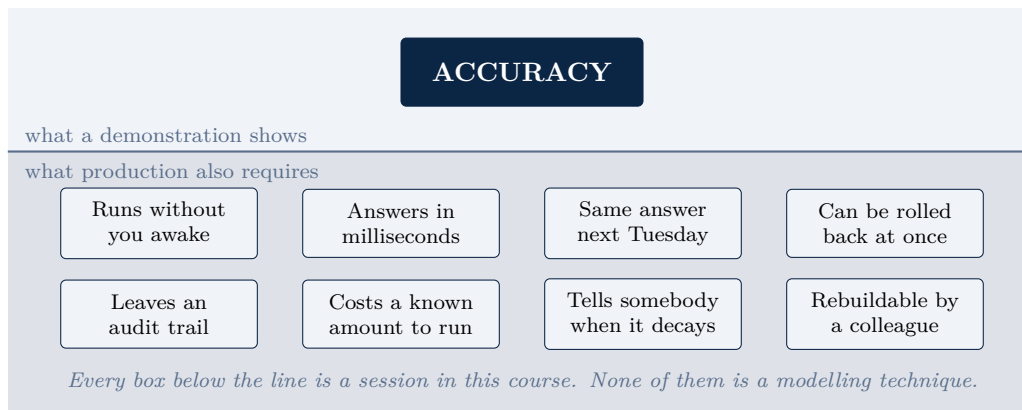


Figure 2: Accuracy is the tip. A brilliant model that nobody can reach, trust or reproduce is not a production system, and the demands below the line sit on top of accuracy rather than instead of it.

4 So what is MLOps

MLOps is the set of practices that takes a model out of a notebook and keeps it running, reliably, for real users. Three properties define it, and every tool in this course exists to deliver one of them.



Figure 3: The three properties, and the part of the course toolchain that delivers each. Notice that none of the three mentions accuracy.

The quiet failure, which is the reason monitoring exists

A web server tells you when it crashes. The alert fires and somebody's phone rings at two in the morning. Loud and immediate.

A model does not tell you when it starts being wrong. It keeps answering, confidently, while the uptime dashboard stays green the entire time. Day one accurate, day thirty slightly off, day sixty often wrong, day ninety wrong more often than not, and the server was never once down.

5 Where machine learning projects fail after the notebook stage

This is the discussion question named in the session plan. The honest answer is that projects die at four distinct points, and only the first is about modelling.

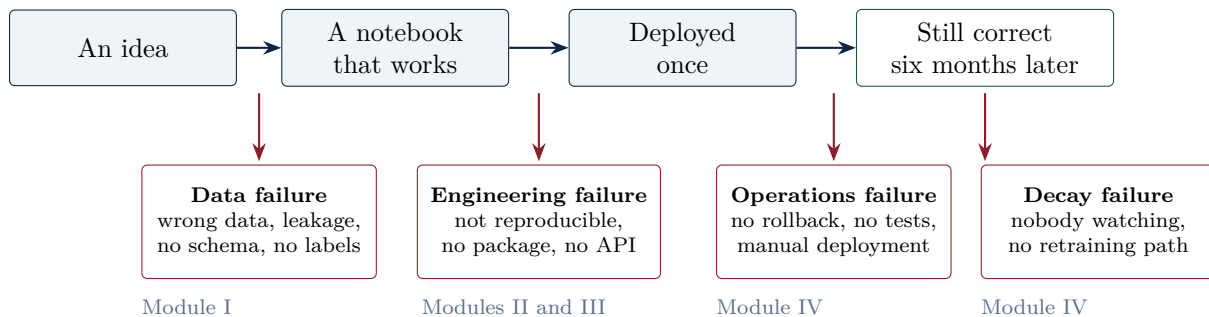


Figure 4: Four exits, and the module of this syllabus that closes each. A project that survives all four is what the industry means by production.

This is a studied question, not folklore

If you want the evidence rather than the anecdote, Paleyes, Urma and Lawrence surveyed published industrial case studies and organised the reported problems by lifecycle stage, in *Challenges in Deploying Machine Learning: a Survey of Case Studies*. It is free at <https://arxiv.org/abs/2011.09926> and it is the single best first paper for this question. Day 3 of the week plan in Part V is built on it.

6 Four things that actually happened

The following are documented cases, each drawn from a primary or peer reviewed source. Know them precisely, because vague versions circulate widely and the details are where the lesson lives.

Case 1 • Zillow Offers, 2021: when the world moves faster than the model

Zillow used an automated valuation model to decide what to pay for houses, buy them, and resell them. In its 2021 annual report to the United States Securities and Exchange Commission, the company recorded an inventory write down of \$407.9 million for the year, stating the cause as *unintentionally purchasing homes at higher prices than the company's current estimates of the future selling prices after selling costs*. The board decided to wind down Zillow Offers in November 2021, a decision the filing attributes to home pricing unpredictability and capacity constraints, and which resulted in roughly a twenty five percent reduction in the company's workforce.

The MLOps failure mode. A model trained on one market regime, deployed into another, whose own purchasing decisions influenced the market it was predicting. Concept drift combined with a feedback loop.

Where this course addresses it. Drift detection and monitoring, Module IV. The production readiness audit, Module IV.

Source: Zillow Group Inc., Form 10-K for fiscal year 2021. <https://www.sec.gov/Archives/edgar/data/1617640/000161764022000013/z-20211231.htm>

Case 2 • Amazon's recruiting engine, 2014 to 2017: when the data carries the bias

Amazon built an experimental tool to score job applicants from one to five stars. Reuters reported in October 2018 that the system was trained on ten years of submitted resumes, which in a male dominated industry came mostly from men. The model learned that male candidates were preferable: it penalised resumes containing the word for women's, and downgraded graduates of two all women colleges. Amazon edited the program to be neutral to those specific terms, but could not establish that it was neutral in every other respect, and the project was abandoned.

The MLOps failure mode. The target variable encoded a historical human decision, so the model learned the historical pattern rather than the intended concept. No line of code was wrong.

Where this course addresses it. Data foundations and data cascades, Module I. Slice based evaluation and the bias audit, Module IV.

Source: J. Dastin, *Amazon scraps secret AI recruiting tool that showed bias against women*, Reuters, 10 October 2018.

Case 3 • Google Flu Trends, 2013: when the input distribution changes underneath you

Google Flu Trends estimated influenza activity from search queries and was widely held up as an exemplary use of large scale data. In February 2013 it was reported to be predicting more than double the proportion of doctor visits for influenza like illness compared with the United States Centers for Disease Control and Prevention, the very quantity it had been built to predict. Lazer and colleagues analysed the failure in *Science* in 2014 and concluded that the large errors were largely avoidable.

The MLOps failure mode. The relationship between the inputs and the target changed over time, partly because search behaviour itself changed, including in response to media coverage. The model was never wrong on the day it was built.

Where this course addresses it. Concept drift against data drift, Module IV.

Source: D. Lazer, R. Kennedy, G. King and A. Vespignani, *The Parable of Google Flu: Traps in Big Data Analysis*, *Science* 343(6176), 1203 to 1205, 2014. <https://doi.org/10.1126/science.1248506>

Case 4 • COVID imaging models, 2020: when nothing is reproducible

During 2020, hundreds of machine learning models were published for detecting or prognosticating COVID-19 from chest radiographs and computed tomography scans. A systematic review in *Nature Machine Intelligence* searched the literature for 1 January to 3 October 2020, identified 2,212 studies, retained 415 after initial screening and 62 after quality screening, and concluded that none of the models identified were of potential clinical use, because of methodological flaws or underlying biases.

The MLOps failure mode. Data leakage, absent external validation, duplicated patients across splits, and undocumented pipelines. Reproducibility failures, not modelling failures.

Where this course addresses it. Reproducible environments and versioned data, Module I. Honest train and test methodology, Practicals 01 and 03.

Source: M. Roberts et al., *Nature Machine Intelligence* 3, 199 to 217, 2021. <https://www.nature.com/articles/s42256-021-00307-0>

FOUR CASES, FOUR DIFFERENT FAILURE MODES, ZERO MODELLING MISTAKES

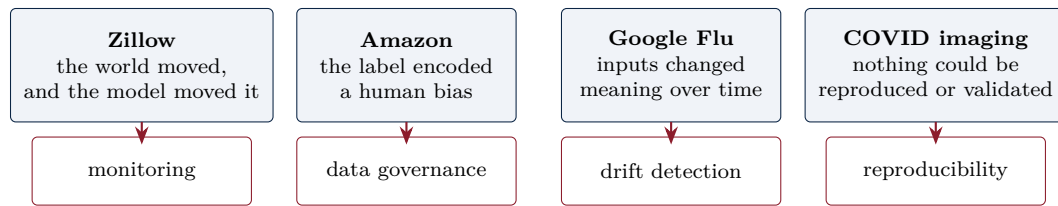


Figure 5: The same conclusion four times over. In each case the mathematics was competent and the engineering around it was not, which is exactly the territory of this course.

7 Common errors in how beginners think about this

The belief	Why it fails
My accuracy is 95 percent, so the work is done	Accuracy is the entry ticket. All four cases above had competent accuracy on the day of release
I will deploy it at the end	Deployment constraints shape the data, the features and the model family from day one. Retrofitting costs more than designing for it
MLOps is just DevOps with a new name	Data and models change on their own; code does not. See Part II
This only matters for large companies	A two person startup feels it fastest, because nobody there has time to babysit a notebook by hand
Monitoring means checking the server is up	The server being up is exactly what makes a decaying model invisible
A better algorithm would have prevented these failures	Google's own engineering guidance opens by advising you to keep the first model simple and get the infrastructure right

8 The running project methodology

This course does not use a new dataset every week. There is one project, a delivery time predictor, and every laboratory moves it one step to the right. Nothing is thrown away, and the whole semester is a single narrative from notebook to production.

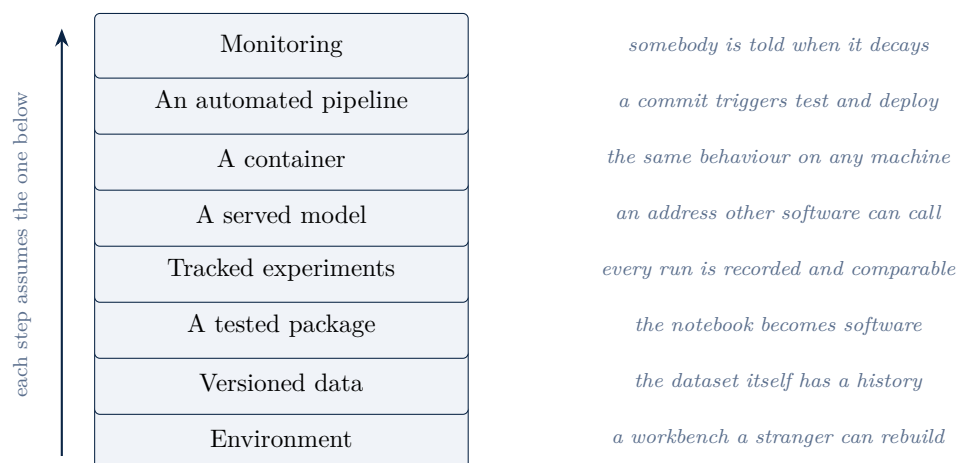


Figure 6: The arc of the semester. By the end you will have a public address a stranger can call, backed by a repository with an automated pipeline and a drift report. This is the running project methodology named in the session plan.

Two consequences follow, and they are worth stating before anybody complains about them later. First, missing a laboratory is expensive, because the next one begins where the last one ended. Second, the work compounds: by week ten you are not starting anything, you are extending something you already understand.

PART II

History, Lifecycle and Roles

9 A short history of the field

Two decades ago, developers wrote code and handed it to a separate operations team who ran it and took the emergency calls. Releases were large, rare and risky. DevOps was the response: one team owning build and run, automation instead of a handover, small frequent releases instead of large rare ones.

Machine learning arrived into that settled world and broke one assumption: that the artefact you ship is written by a human and therefore reviewable line by line.

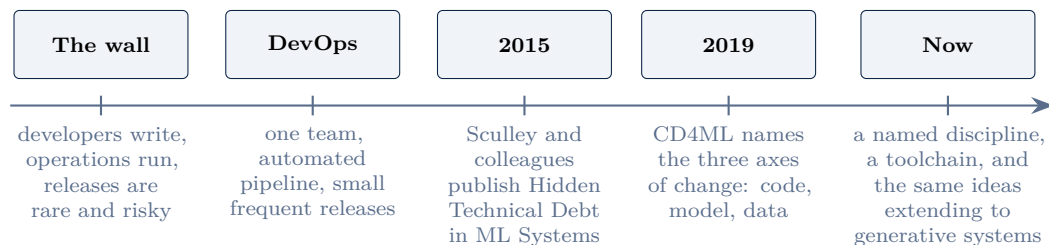
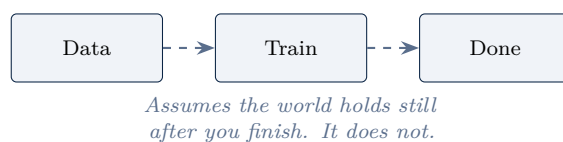


Figure 7: The lineage of the field. MLOps did not appear from nowhere: it is DevOps discipline applied to artefacts that are compiled from data rather than written by hand.

Two documents did most of the intellectual work, and both are free. The 2015 NeurIPS paper by Sculley and colleagues argued that machine learning systems accumulate a distinctive kind of technical debt, and named the entanglement property, changing anything changes everything. The 2019 article by Sato, Wider and Windheuser then framed the engineering answer: machine learning applications are subject to change along three axes, the code, the model and the data, and continuous delivery has to be extended to cover all three. Read both this week; they are Days 2 and 4 of the plan in Part V.

10 A line, or a loop

WHAT STUDENTS PICTURE



WHAT ACTUALLY HAPPENS

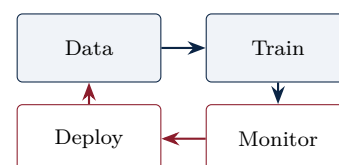


Figure 8: The straight line is not a simplification of the loop, it is a different claim about the world. The loop says that what you learn from a live model becomes next month's training data.

11 The six stages

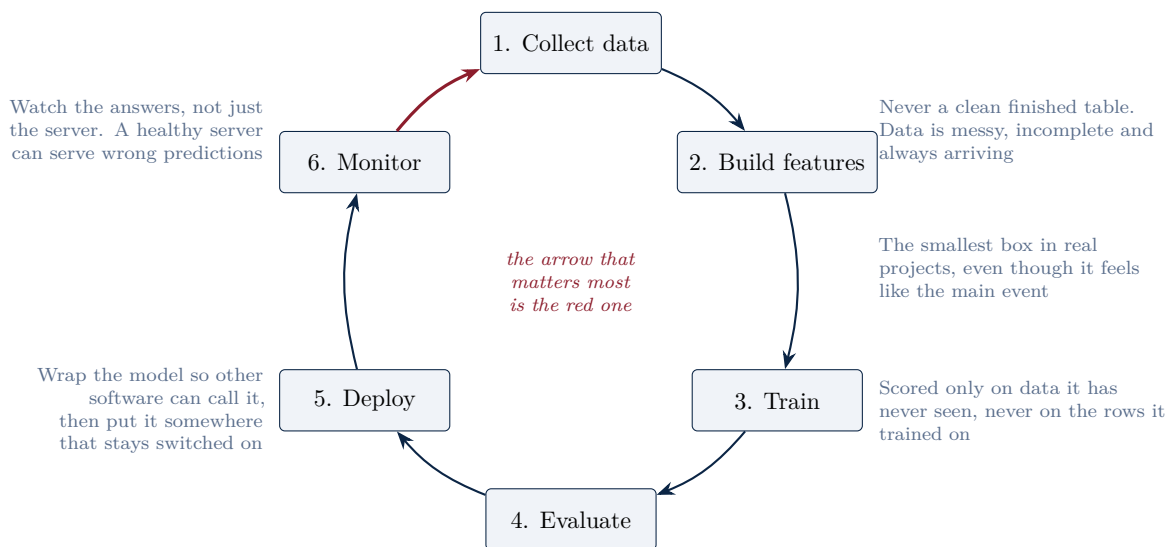


Figure 9: The production machine learning lifecycle. Training, the stage that occupies almost all of a normal machine learning course, is one sixth of the circle and typically the smallest part of the engineering effort.

11.1 What happens, what goes wrong, and which tool answers it

Stage	On our delivery project	What goes wrong	Tool, later
1. Collect data	Orders, distances, timestamps, rider identifiers, weather	Silent schema changes, missing labels, duplicated rows across splits	DVC, schema validation
2. Build features	Distance in kilometres, hour of day, kitchen load, is it raining	Transformations that exist in the notebook but not at serving time	Configuration files, shared transformation code
3. Train	A regression baseline, then a better model	Untracked runs, no seed, no record of which data produced which model	MLflow
4. Evaluate	Mean absolute error on held out orders, and on each city separately	Aggregate metrics hiding a failing subgroup	Slice based evaluation
5. Deploy	A prediction endpoint the ordering application can call	Manual deployment, no rollback path, environment differs from training	FastAPI, Docker, GitHub Actions
6. Monitor	Watch input distributions and prediction quality by city	Nobody watching, or watching only uptime	Evidently, drift logged to MLflow

The one sentence version

Stage three is where the intelligence lives. Stages one, two, five and six are where the projects die.

12 The six stage loop is a teaching simplification

Six stages is the right picture to start from, and it is not what a real team's process diagram looks like. The most cited empirical account is Amershi and colleagues at Microsoft Research, who observed software teams building machine learning applications and described a nine stage workflow: model requirements, data collection, data cleaning, data labelling, feature engineering, model training, model evaluation, model deployment, and model monitoring.

Two details from that paper are worth carrying into every later session.

1. **The feedback arrows are not only from the end.** Evaluation and monitoring may loop back to *any* earlier stage, and training may loop back to feature engineering. The process is not a circle so much as a graph with many return paths.
2. **Four of the nine stages are about data, not models.** Collection, cleaning, labelling and feature engineering all sit upstream of anything you would recognise as machine learning.

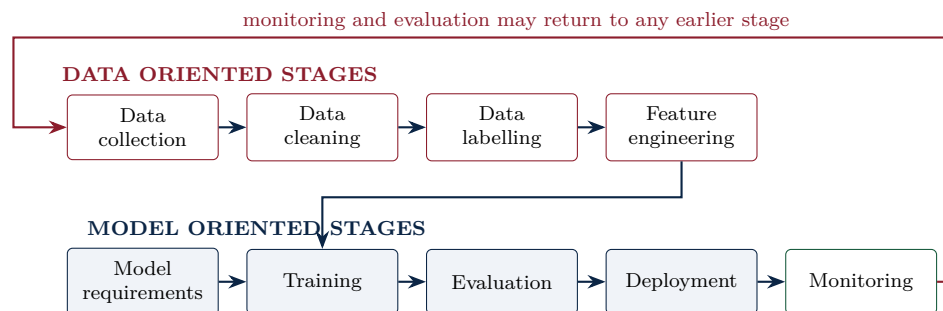
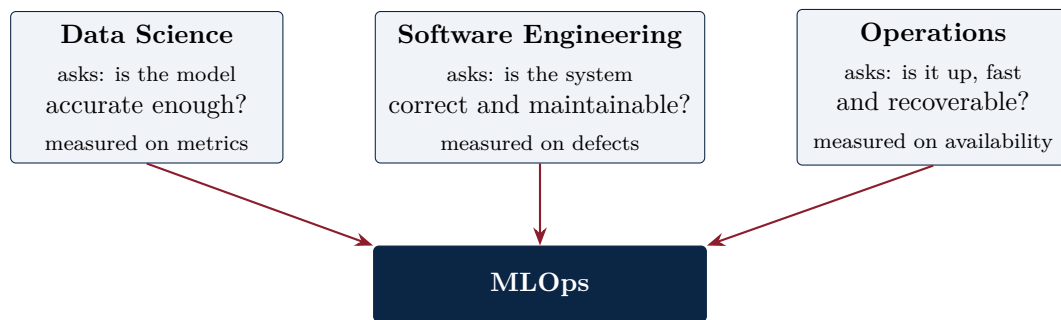


Figure 10: The nine stage workflow observed at Microsoft, redrawn. Four stages are data oriented and five are model oriented, and the return arrows reach much further back than the tidy circle suggests.

13 Bridging Data Science, DevOps and Engineering

MLOps exists at the meeting point of three roles that were trained separately, use different vocabulary, and are measured on different things. Most real failures happen at these boundaries rather than inside any one of them.



The discipline is the shared vocabulary and the shared pipeline that let these three answer one question together: can we ship this, and will we know when it stops working?

Figure 11: Three roles, three definitions of success. A model can be accurate, well engineered and highly available while still being useless, if the three were optimised separately.

The Microsoft study identified three ways in which building machine learning applications differs fundamentally from prior software work, and each maps onto a boundary in the figure above.

The difference	Why it strains the boundary
Discovering, managing and versioning data is far more complex than in other software engineering	The data scientist treats the dataset as given; the engineer has no tool in the traditional toolbox that versions a five gigabyte file sensibly. This is why DVC exists, and why it is Module I
Model customisation and reuse require very different skills from those usually found in software teams	A team can be excellent at software and still be unable to judge whether a model should ship. This is why the oral assessment in this course asks you to justify, not just to run
Machine learning components are harder to keep as distinct modules, because models may be entangled and experience non monotonic error behaviour	Improving one part can make the whole worse, which breaks the assumption that modular testing is sufficient

What this means for you personally

You are being trained to be the person who can hold all three conversations. That person is rare, is not produced by a modelling course alone, and is what the job titles machine learning engineer and platform engineer actually describe.

14 From DevOps to MLOps

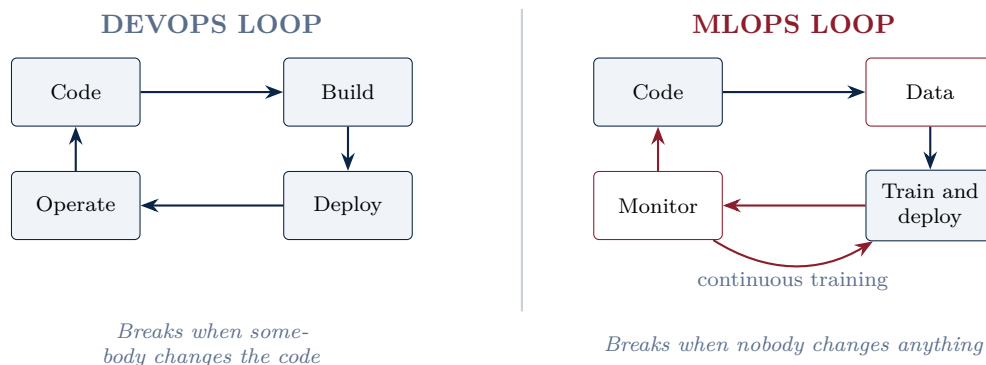


Figure 12: The extra arrow is the whole difference. Continuous training has no counterpart in DevOps, because software does not decay on its own when the world moves.

14.1 What carries over unchanged

1. **Version everything.** If it affects the output, it has a version number.
2. **Automate the path to release.** A human doing the same steps by hand will eventually do them differently.
3. **Small, frequent changes.** A small change that breaks something is easy to find. A large one is not.
4. **If a manual step hurts, script it.** Pain is a signal about missing automation, not a reason for heroism.

14.2 What is added

	DevOps	MLOps
What you version	Code	Code, data and model
What tests check	Does the code work	Is the model still accurate
Why it breaks	Someone changed the code	The world changed
What you roll back to	The previous build	The previous model version, and the transformations that matched it
What triggers a release	A commit	A commit, new data, or a drift signal
What you ship	A binary	A model, a serving contract, and a monitoring plan

15 Three moving parts, not one

	CODE	DATA	MODEL
Versioned by	Git	DVC	Model registry
Tested by	pytest	Schema checks	Quality gates
Changes when	you edit it	the world moves	either of the other two moves

Figure 13: The three axes of change named in the CD4ML article. Read the bottom row carefully: only the first column changes when a human decides to change it, and that is the entire reason this discipline exists.

PART III

Classroom Activity

16 A Kaggle notebook against a bank's fraud detection service

Both predict something. Both can be described as machine learning. Almost nothing else about them is the same. Complete this grid in pairs, then compare with the row hints on the next page.

Dimension	Kaggle notebook	Bank fraud detection service
Who calls it		
How often		
Latency allowed		
What happens if it is wrong		
Who sees the error		
How the data arrives		
Are labels available		
How long until it is stale		
Who can roll it back		
What must be auditable		
Cost of a false positive		
Cost of a false negative		

16.1 Row hints, for after the attempt

Dimension	The distinction worth drawing out
Latency allowed	A notebook may take an hour. A card authorisation decision has to return inside the payment flow, and a slow answer is a failed answer
Are labels available	Kaggle hands you every label. A bank learns whether a transaction was fraudulent when the customer disputes it, possibly weeks later. This single difference reshapes the entire monitoring design
How long until it is stale	A competition dataset is frozen. Fraud tactics adapt deliberately and adversarially, which is concept drift with an opponent

Dimension	The distinction worth drawing out
Cost of a false positive	A wrong Kaggle prediction costs a place on a leaderboard. A blocked legitimate card costs a customer, a support call, and possibly a regulator's attention
What must be auditable	Nothing, in a competition. In a bank, the reason for a decision may have to be produced years later, which is why lineage is not paperwork
Who can roll it back	Nobody needs to. In a bank, somebody must be able to, in minutes, at three in the morning

Closing question for the activity

Which single row would you most want to get right before writing any modelling code? Most teams answer with the labels row, and that is the correct instinct: if you cannot measure whether you were right, nothing downstream can be trusted.

PART IV

Consolidation

17 Discussion question for this session

Where do machine learning projects fail after the notebook stage?

Work in pairs. Choose one of the four documented cases in Section 6, or a failure you have read about elsewhere. Identify, in writing:

1. Which of the four exits in Figure 4 the project took.
2. Which single practice, from the three properties in Figure 3, would most likely have caught it.
3. What evidence would have made the failure visible earlier, and who would have had to look at it.

18 Check your understanding

1. Give three concrete reasons a model that scores well in a notebook might be useless to a company on Monday morning.
2. State the three properties that define MLOps, and name one tool from this course that delivers each.
3. Explain why a green uptime dashboard is compatible with a model that is wrong most of the time.
4. In the lifecycle loop, which arrow do students usually omit, and what does omitting it assume about the world?
5. Name two things that MLOps versions which DevOps does not.
6. Why can a machine learning system break when nobody has changed any code?
7. In the Zillow case, why is this described as a feedback loop rather than only drift?
8. In the Amazon case, no line of code was wrong. Where did the error actually live?
9. Google Flu Trends was accurate when it was built. Which stage of the lifecycle should have caught its decay?
10. The COVID imaging review examined 62 studies and found none clinically usable. Which single property from Figure 3 was missing across the field?
11. Four of the nine stages in the Microsoft workflow are data oriented. What does that imply about where you should expect to spend your semester?
12. Name the three roles MLOps sits between, and give one thing each of them is measured on.
13. Why does this course insist on one running project rather than a new dataset each week?
14. Write the one sentence you would use to explain MLOps to a classmate who has never heard the term.

19 Vocabulary introduced today

Term	Meaning
MLOps	The practices that take a model out of a notebook and keep it running reliably for real users
Production	An environment where software serves real users continuously, without its author present
Reproducible	The same code and data give the same result on any machine on any day
Reachable	Other software can call the model over a network address
Watched	Degradation triggers an automatic signal to a human
Data drift	The distribution of the inputs changes over time
Concept drift	The relationship between inputs and the target changes over time
Feedback loop	The model's own predictions influence the world that generates its next training data
Data leakage	Information unavailable at prediction time contaminates training, inflating offline scores
Training and serving skew	Transformations differ between training and inference
Model artefact	The serialised model file, plus the metadata needed to use and trace it
Continuous training	Automated retraining triggered by a schedule, new data or a drift signal
Lineage	The traceable chain from a prediction back to the data and commit that produced it
CACE	Changing anything changes everything: the entanglement property of learned systems
Technical debt	Engineering shortcuts that cost more to carry than to fix, and which machine learning systems accumulate in distinctive forms

PART V

The Week Ahead

20 Seven days, about eight hours in total

This is not optional reading dressed up as a plan. Each day names one resource, the time it takes, and one thing you produce. Keep the outputs in a single file; several of them are reusable in the Self Learning Assignment later in the semester.

Day 1 • Ground yourself in the practice • 60 minutes

Read. Google's *Rules of Machine Learning: Best Practices for ML Engineering* by Martin Zinkevich. Read the overview and Rules 1 to 12 only. <https://developers.google.com/machine-learning/guides/rules-of-ml>

Why. It is the most quoted piece of practical engineering guidance in the field, written from inside Google, and it argues for infrastructure before sophistication. Rule 4, keep the first model simple and get the infrastructure right, is the thesis of your entire first month.

Produce. Three rules that contradict how you have worked so far, one line each on why.

Day 2 • The founding paper • 75 minutes

Read. Sculley et al., *Hidden Technical Debt in Machine Learning Systems*, NeurIPS 2015. Nine pages, free. https://proceedings.neurips.cc/paper_files/paper/2015/file/86df7dcfd896fcdf2674f757a2463eba-Paper.pdf

Why. It named the problem the whole discipline answers, including entanglement, correction cascades, undeclared consumers and pipeline jungles. Every serious MLOps talk still cites it.

Produce. A definition in your own words of CACE, and one example of it from any model you have built.

Day 3 • Evidence, not anecdote • 60 minutes

Read. Paleyes, Urma and Lawrence, *Challenges in Deploying Machine Learning: a Survey of Case Studies*. Read the deployment and monitoring sections. <https://arxiv.org/abs/2011.09926>

Why. It organises real reported industrial problems by lifecycle stage, which turns today's discussion question from opinion into a literature backed answer.

Produce. Map three challenges from the survey onto the four exits in Figure 4 of this handout.

Day 4 • The engineering answer • 75 minutes

Read. Sato, Wider and Windheuser, *Continuous Delivery for Machine Learning*. <https://martinfowler.com/articles/cd4ml.html>

Why. It is the clearest single article on the bridge from DevOps to MLOps, it names the three axes of change used in Section 14, and it walks a full worked example rather than a diagram.

Produce. Sketch the CD4ML pipeline for our delivery time predictor, on one page, by hand.

Day 5 • How real teams work • 60 minutes

Read. Amershi et al., *Software Engineering for Machine Learning: A Case Study*, ICSE 2019. Free from Microsoft Research. https://www.microsoft.com/en-us/research/wp-content/uploads/2019/03/amershi-icse-2019_Software_Engineering_for_Machine_Learning.pdf

Why. It is an observational study of actual teams, it gives the nine stage workflow in Section 11, and it names the three ways this work differs from ordinary software engineering.

Produce. Which of the nine stages have you personally ever done? Be honest. The gaps are your semester.

Day 6 • The industry maturity view • 60 minutes

Read. Google Cloud, *MLOps: continuous delivery and automation pipelines in machine learning*. <https://cloud.google.com/architecture/mlops-continuous-delivery-and-automation-pipelines-in-machine-learning>

Optional companion. Databricks, *The Big Book of MLOps*, free. <https://www.databricks.com/wp-content/uploads/2022/09/The-Big-Book-of-MLOps-v6-082322.pdf>

Why. Levels 0, 1 and 2 are the vocabulary every practitioner uses to describe how automated a team actually is. You will be asked this in interviews.

Produce. Place our course project at a level, and say what would have to be true to move up one.

Day 7 • Start the habit • 90 minutes

Watch and do. Week 1 of *Machine Learning in Production* by Andrew Ng, DeepLearning.AI, free to audit. <https://www.coursera.org/learn/introduction-to-machine-learning-in-production>

Then. Open the MLOps Zoomcamp repository and read the module list only, so you know what is coming. <https://github.com/DataTalksClub/mlops-zoomcamp>

Produce. One paragraph: what surprised you most this week, and what you now expect to find hardest.

If you only have one hour all week

Day 1, then Day 4. Rules of Machine Learning gives you the judgement, and the CD4ML article gives you the shape of the pipeline. Everything else in this course hangs on those two.

PART VI

Resource Library

All links verified on 21 August 2026. Prefer these to blog summaries and video digests. Where a paper and a summary both exist, read the paper; it is usually shorter than the summary and always more precise.

21 Tier 1, the primary literature

Source	Where, and why it matters
Sculley et al., <i>Hidden Technical Debt in Machine Learning Systems</i> , NeurIPS 2015	https://proceedings.neurips.cc/paper_files/paper/2015/file/86df7dcfd896fcaf2674f757a2463eba-Paper.pdf Named the problem the field answers
Sambasivan et al., <i>Data Cascades in High Stakes AI</i> , CHI 2021, open access	https://dl.acm.org/doi/10.1145/3411764.3445518 The prescribed reading for Session 13; why upstream data neglect compounds
Amershi et al., <i>Software Engineering for Machine Learning: A Case Study</i> , ICSE 2019	https://www.microsoft.com/en-us/research/wp-content/uploads/2019/03/amershi-icse-2019_Software_Engineering_for_Machine_Learning.pdf The nine stage workflow, from observing real teams
Paleyes, Urma and Lawrence, <i>Challenges in Deploying Machine Learning: a Survey of Case Studies</i>	https://arxiv.org/abs/2011.09926 Reported failures organised by lifecycle stage
Lazer et al., <i>The Parable of Google Flu</i> , Science 343, 2014	https://doi.org/10.1126/science.1248506 The canonical concept drift failure
Roberts et al., <i>Nature Machine Intelligence</i> 3, 2021	https://www.nature.com/articles/s42256-021-00307-0 What happens to a whole field without reproducibility
Bernardi, Mavridis and Estevez, <i>150 successful machine learning models: 6 lessons learned at Booking.com</i> , KDD 2019, pages 1743 to 1751	Industry scale evidence that offline gains and business value are not the same thing

22 Tier 2, canonical practitioner writing

Source	Where, and why it matters
Zinkevich, <i>Rules of Machine Learning</i> , Google	https://developers.google.com/machine-learning/guides/rules-of-ml Forty three rules of practical judgement; also as a PDF at https://martin.zinkevich.org/rules_of_ml/rules_of_ml.pdf
Sato, Wider and Windheuser, <i>Continuous Delivery for Machine Learning</i>	https://martinfowler.com/articles/cd4ml.html The DevOps to MLOps bridge, with a full worked example
Google Cloud, <i>MLOps: continuous delivery and automation pipelines</i>	https://cloud.google.com/architecture/mlops-continuous-delivery-and-automation-pipelines-in-machine-learning Maturity levels 0, 1 and 2
Databricks, <i>The Big Book of MLOps</i> , free	https://www.databricks.com/wp-content/uploads/2022/09/The-Big-Book-of-MLOps-v6-082322.pdf Deployment patterns and reference architectures
Google, <i>ML Universal Guides</i>	https://developers.google.com/machine-learning/guides Data preparation, evaluation and fairness guides in the same style

23 Tier 3, full courses, all free

Course	Where
Machine Learning in Production, DeepLearning.AI, free to audit	https://www.coursera.org/learn/introduction-to-machine-learning-in-production
MLOps Zoomcamp, DataTalks.Club, open source and project based	https://github.com/DataTalksClub/mlops-zoomcamp
Made With ML, Goku Mohandas	https://madewithml.com and https://github.com/GokuMohandas/mlops-course
Stanford CS 329S, Machine Learning Systems Design, open notes	https://stanford-cs329s.github.io
NPTEL Cloud Computing, IIT Kharagpur	https://nptel.ac.in/courses/106105167

24 Tier 4, official documentation for the course toolchain

The toolchain changes faster than any book, so the manual is the only source that is current by definition. Learning to work from it is an explicit outcome of this course, mapped to PO12.

Tool	Official manual
Python virtual environments	https://docs.python.org/3/library/venv.html
pip requirements format	https://pip.pypa.io/en/stable/reference/requirements-file-format/
Git, the Pro Git book, free	https://git-scm.com/book/en/v2
DVC	https://doc.dvc.org/start
scikit-learn	https://scikit-learn.org/stable/user_guide.html
pytest	https://docs.pytest.org/en/stable/getting-started.html
MLflow	https://mlflow.org/docs/latest/ml/getting-started/
FastAPI	https://fastapi.tiangolo.com
Docker	https://docs.docker.com/get-started/
Kubernetes concepts	https://kubernetes.io/docs/concepts/
GitHub Actions	https://docs.github.com/en/actions/get-started/quickstart
Apache Airflow	https://airflow.apache.org/docs/apache-airflow/stable/start.html
Apache Kafka	https://kafka.apache.org/documentation/
Evidently	https://docs.evidentlyai.com/quickstart_ml
Grafana	https://grafana.com/docs/

Everything you have been taught so far assumes the world holds still while you finish your work.

It does not.

This course is about what you build so that your work survives the world moving.

Next lecture. Lecture 02: the running project stated plainly, the eight tools with one job each, and setting up your environment before the first practical.